Neutral Change is a Characteristic Property of Clusterized Language Communities

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Question

- Can language change be **neutral**?
- I.e., not motivated by **biases**, which could be
  - perceptual or articulatory (Ohala, 1989; Pierrehumbert, 2001)
  - cognitive (Lightfoot, 1979; Roberts and Roussou, 2003)
  - social (Labov, 1972)
  - distributional (Niyogi and Berwick, 1997; Yang, 2000)
  - hermeneutic (language as a teleological system; 'preferred' vs. 'dispreferred' states; Itkonen, 1982; Vennemann, 1993)
- This neutrality hypothesis has been suggested before (Lass, 1997; Trudgill, 2008)

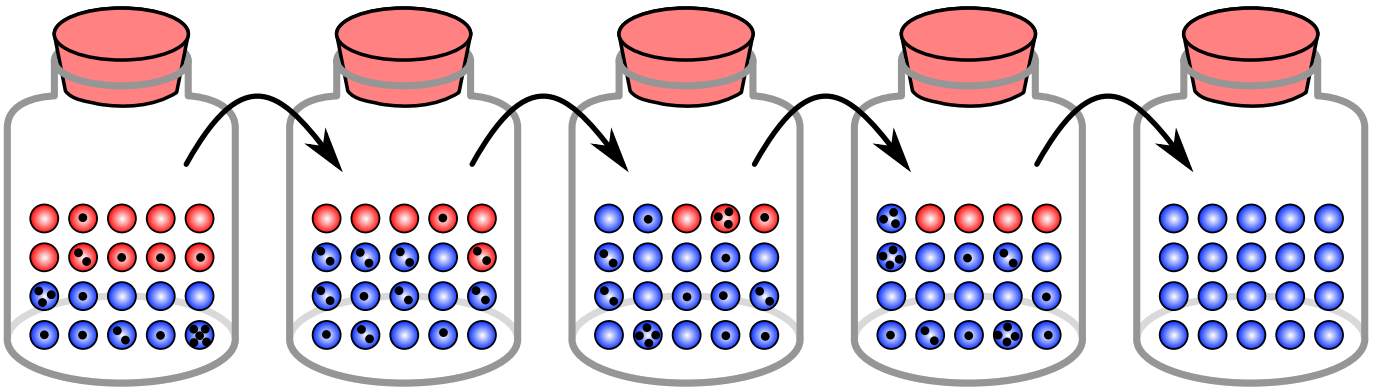
Strategy

- Define a mathematical model of language change that works on neutral principles
- Analyse behaviour of model, either mathematically or running numerical simulations
- For this purpose, a notion of 'well-behavedness' of change will be defined and operationalized mathematically
- See if neutral change can occur, and whether it is a robust phenomenon or a chance occurrence

Neutrality

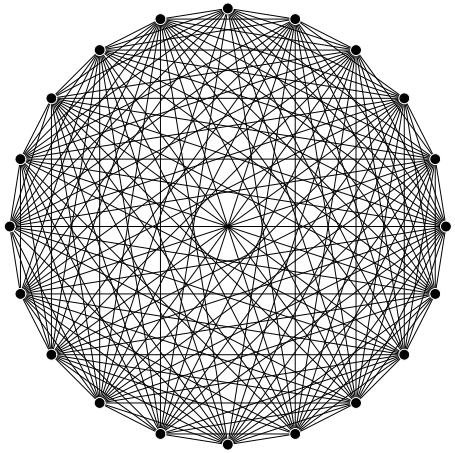
Definition (Neutral change)

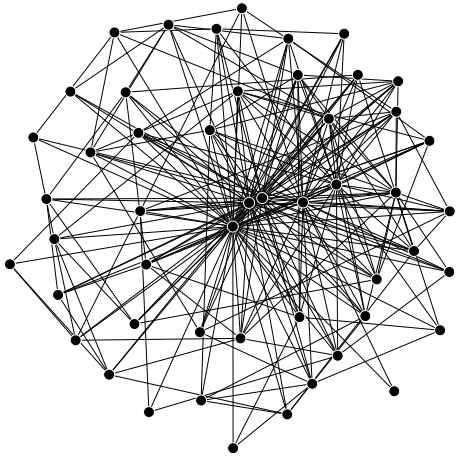
The probability with which a speaker acquires a certain linguistic variant out of a number of competing variants equals the relative frequency of that variant in the speaker's linguistic neighbourhood, *modulo* a small probability of innovating, uniformly at random, another variant from among all biologically possible variants.



Neutral change?

	static	dynamic
well-mixing		
structured		





Neutral change?

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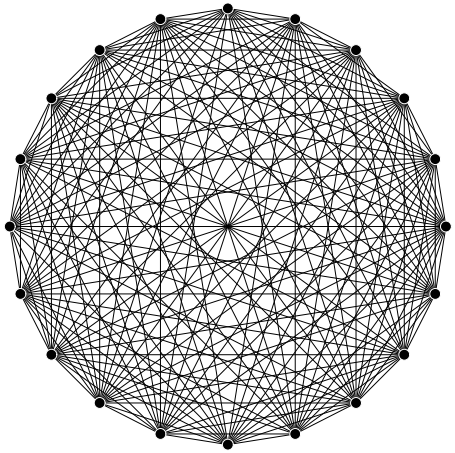

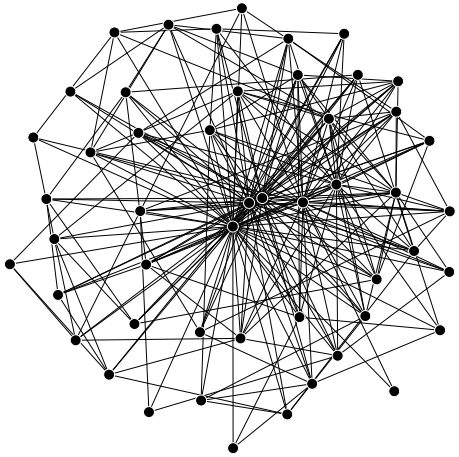

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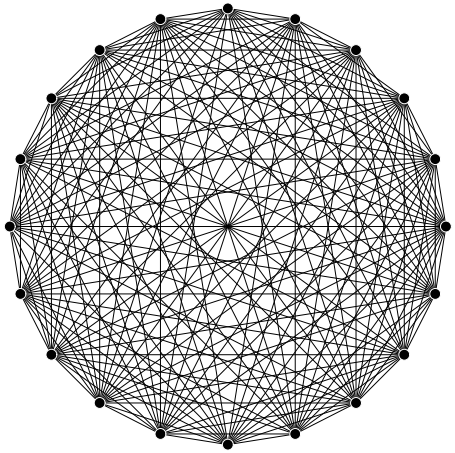

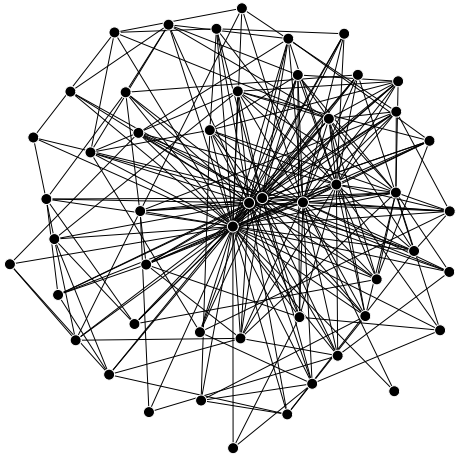

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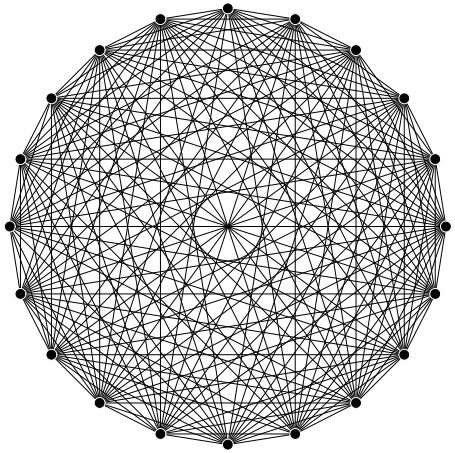
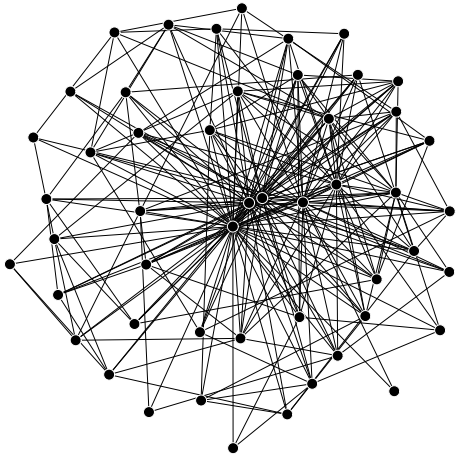
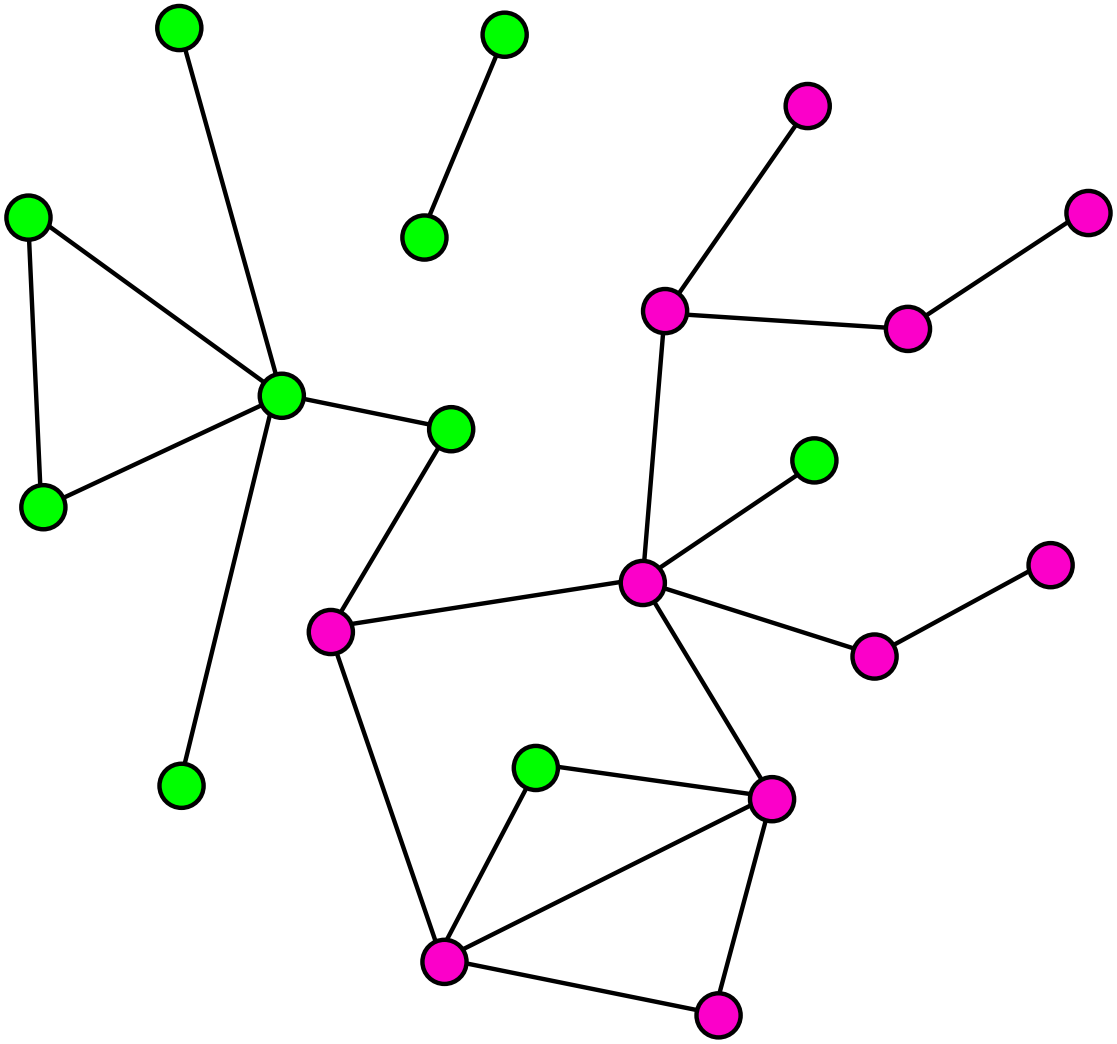
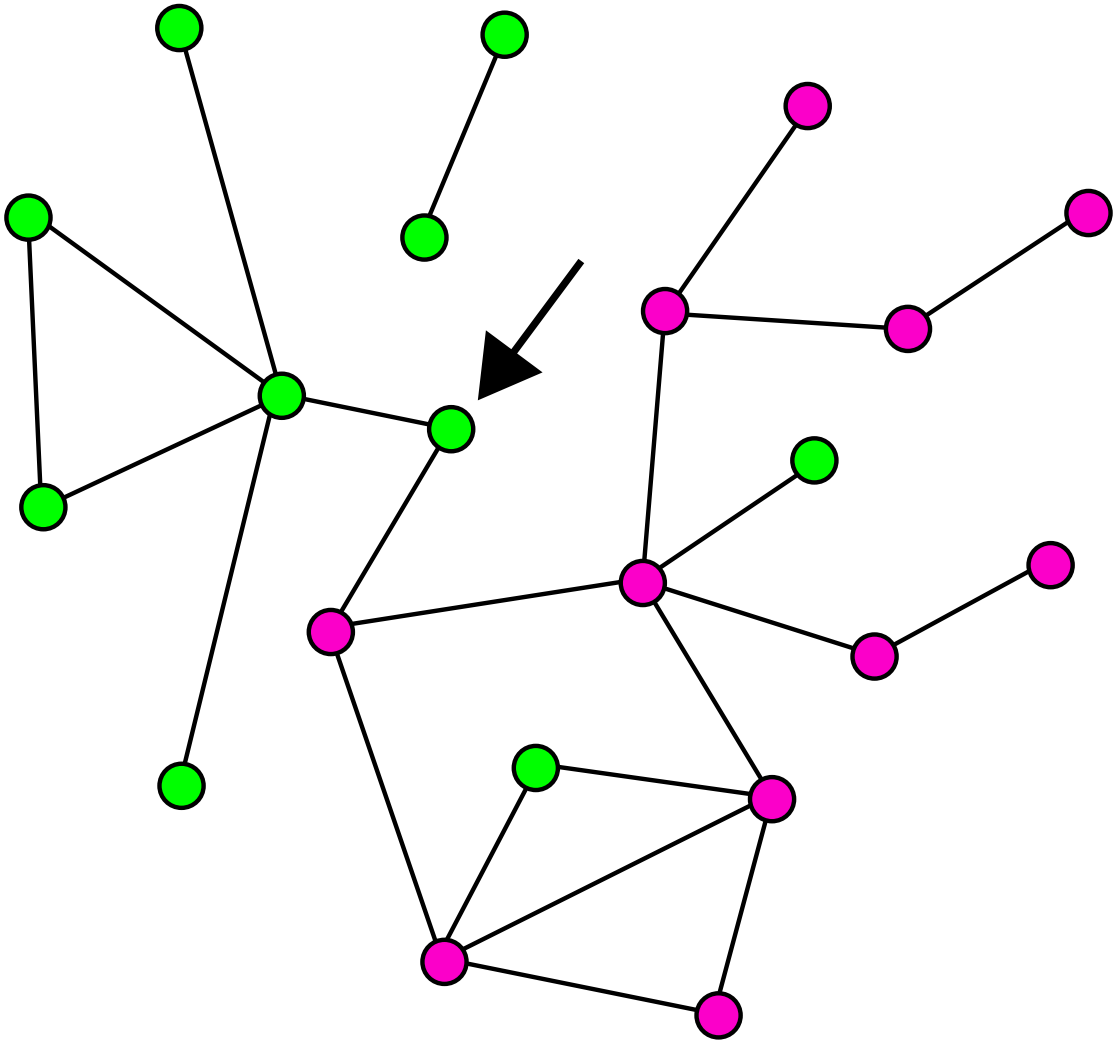


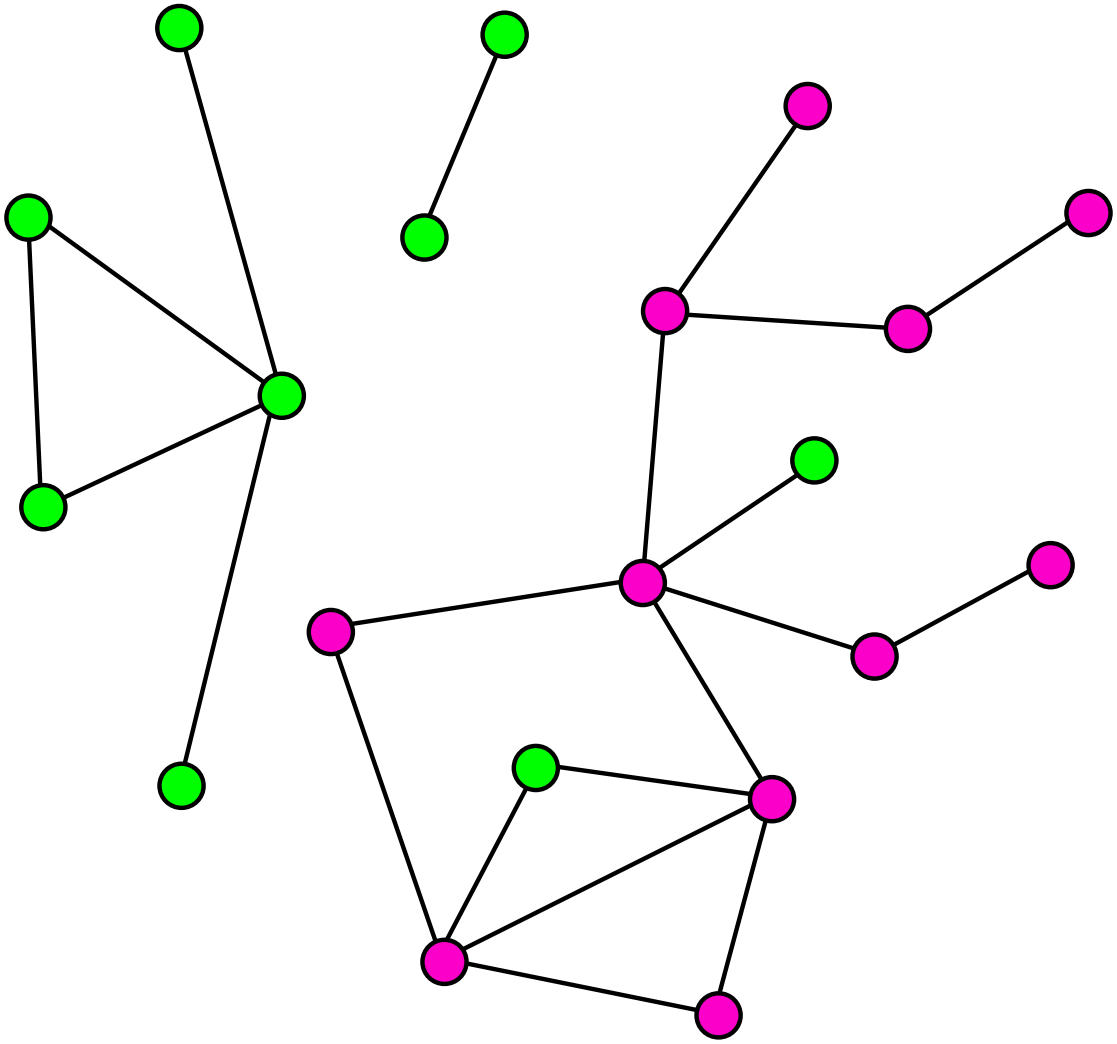


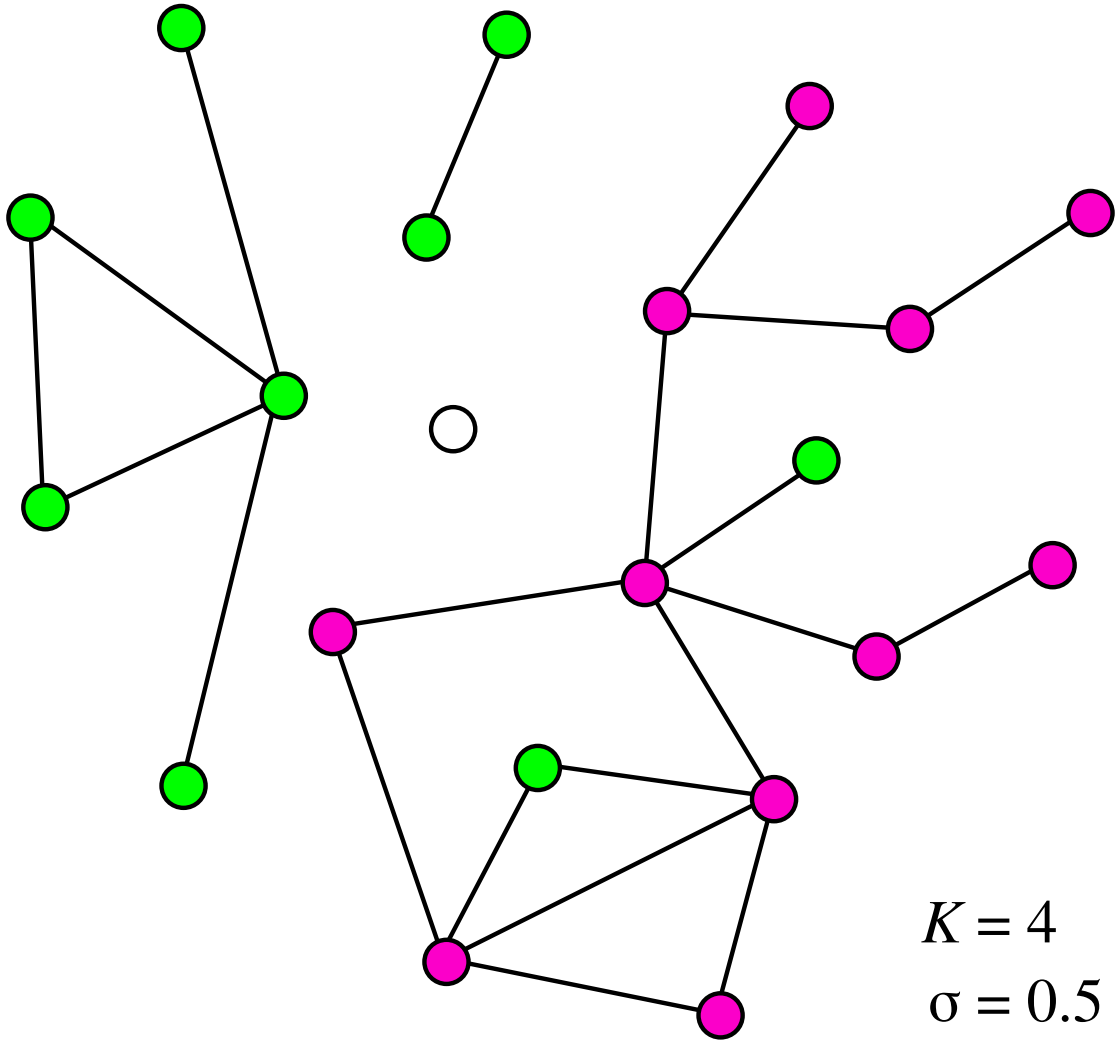
Model

- Social network of N speakers
- Each speaker monolingual and fixed in one of C possible variants
- Network-rewiring: at each time step, one speaker is removed (uniformly at random) and replaced with a new one who is connected to K speakers as follows:
  - With probability σ, make connection to the best-connected speakers in the network
  - With probability $1 - \sigma$, make connection uniformly at random
  - Repeat K times; this defines the speaker's **neighbourhood**
- Frequency-dependent variant adoption: speaker now adopts variant X with probability p, where p = frequency of X in speaker's neighbourhood...
- ...except that with (low) probability μ, he innovates (again, uniformly at random)





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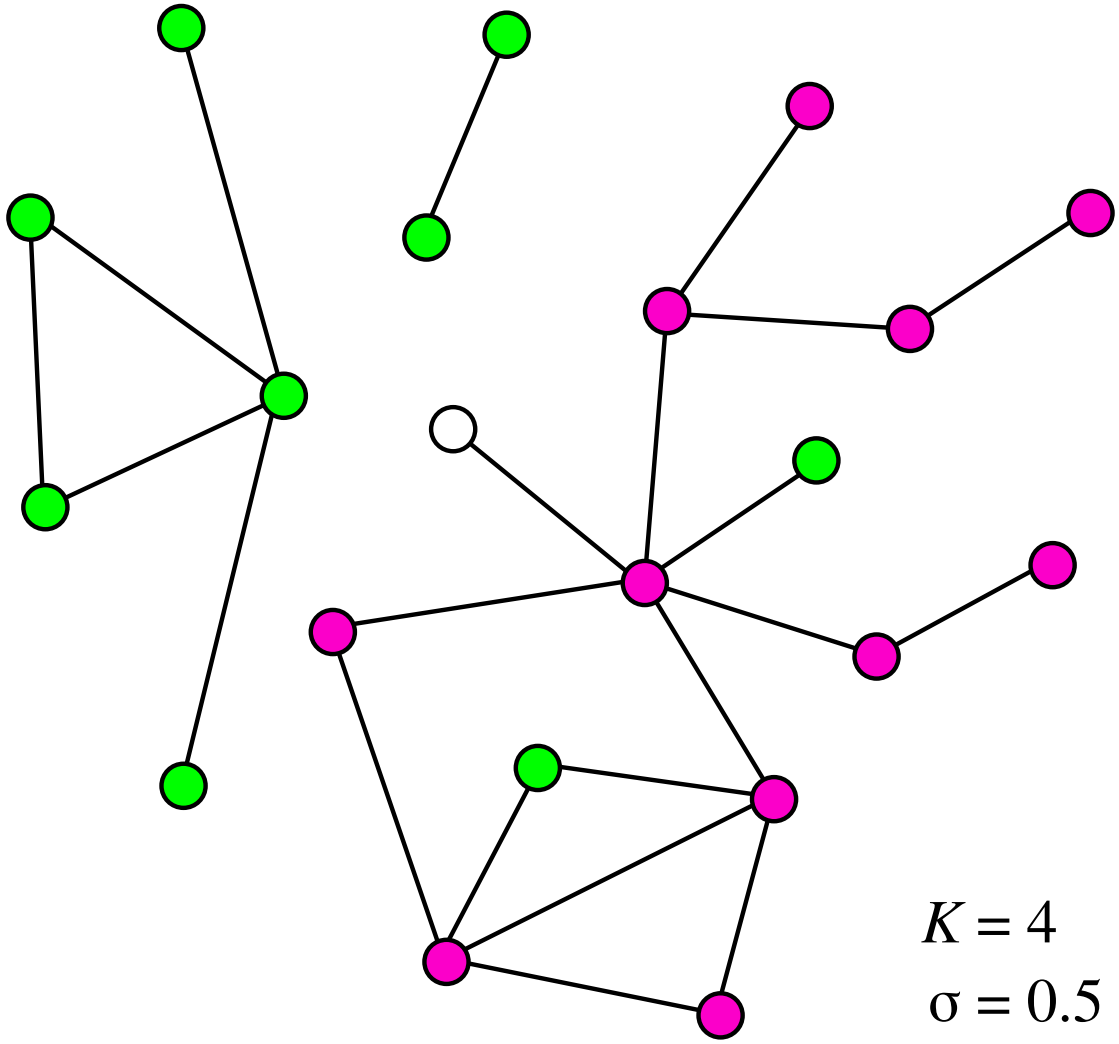
$\sigma = 0.5$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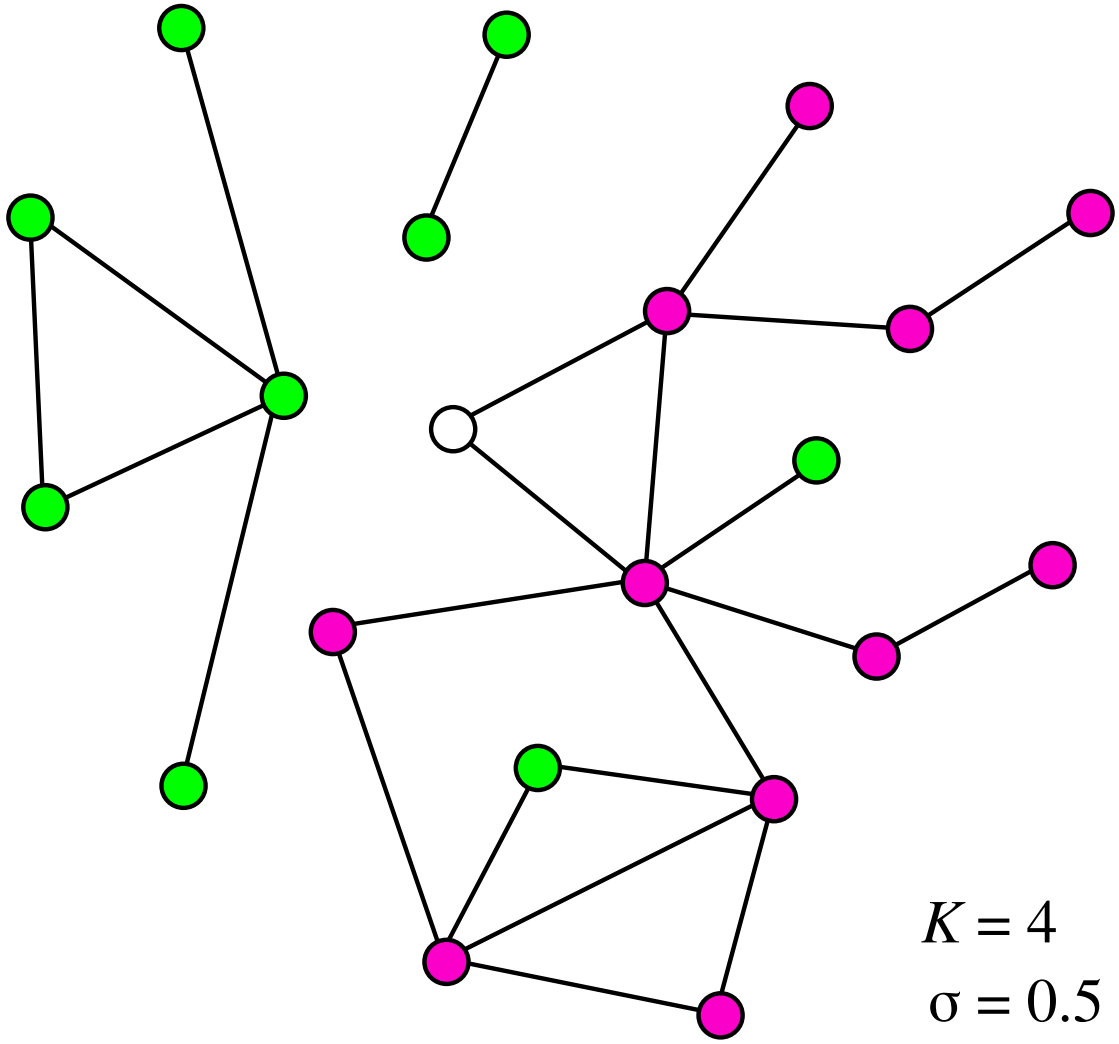
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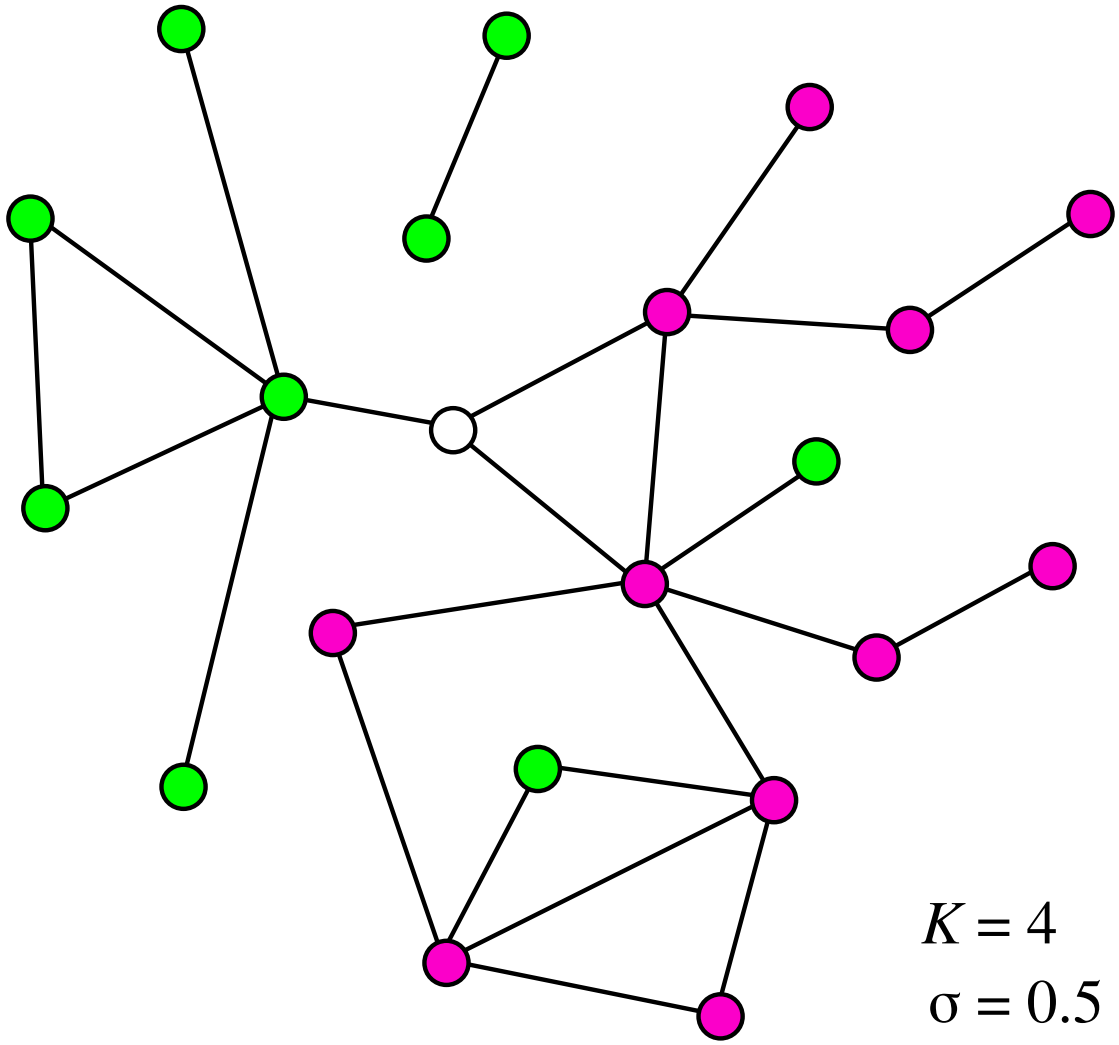
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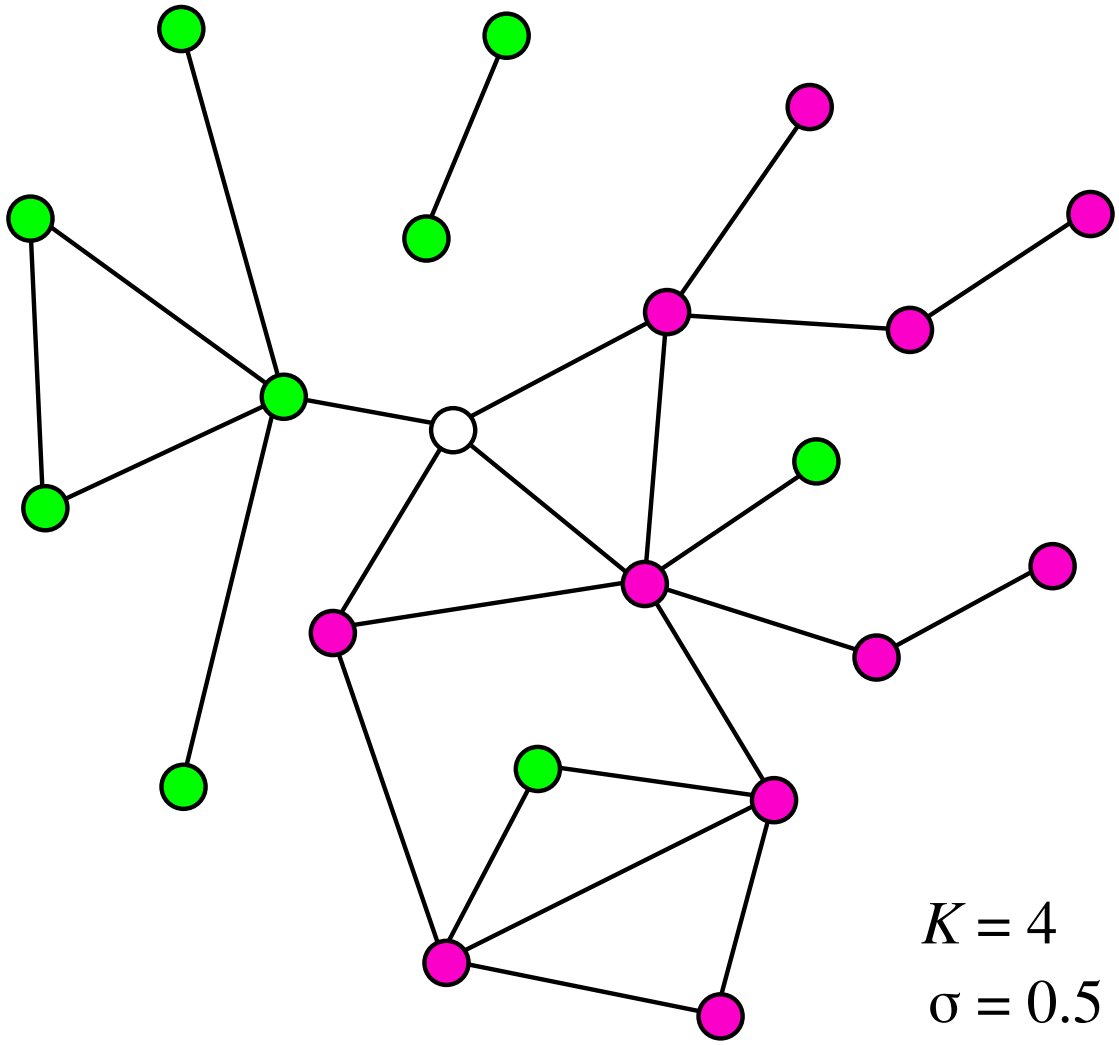
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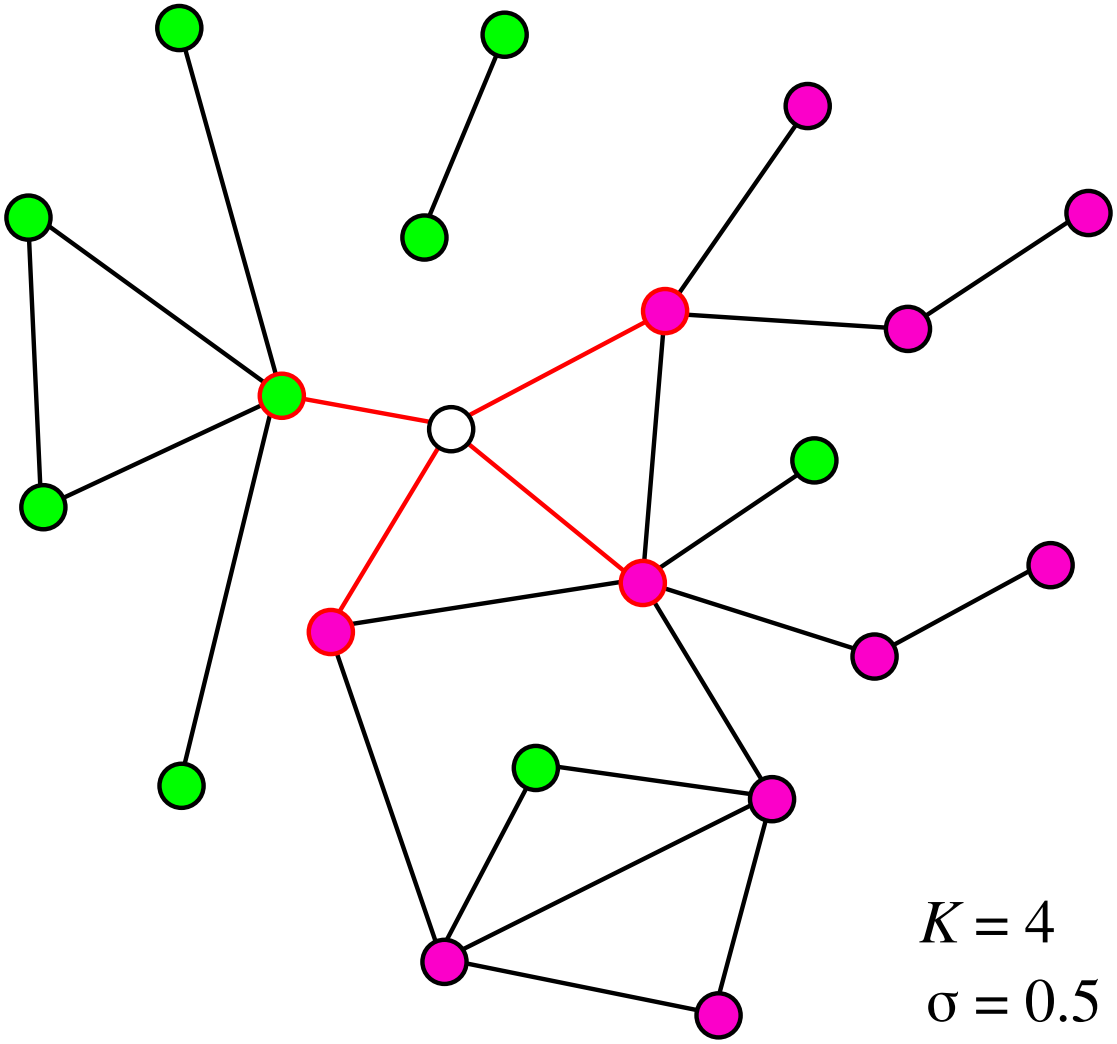
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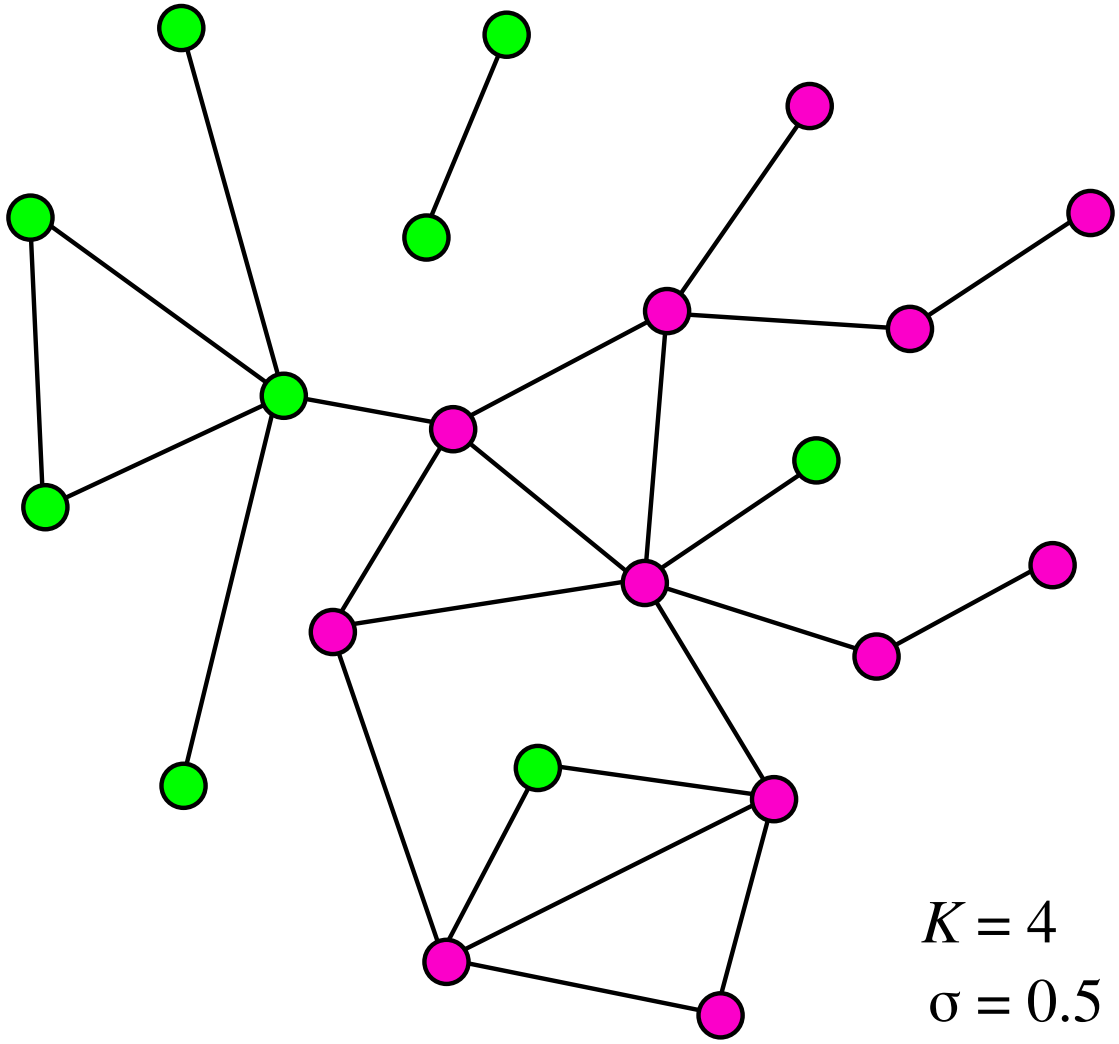
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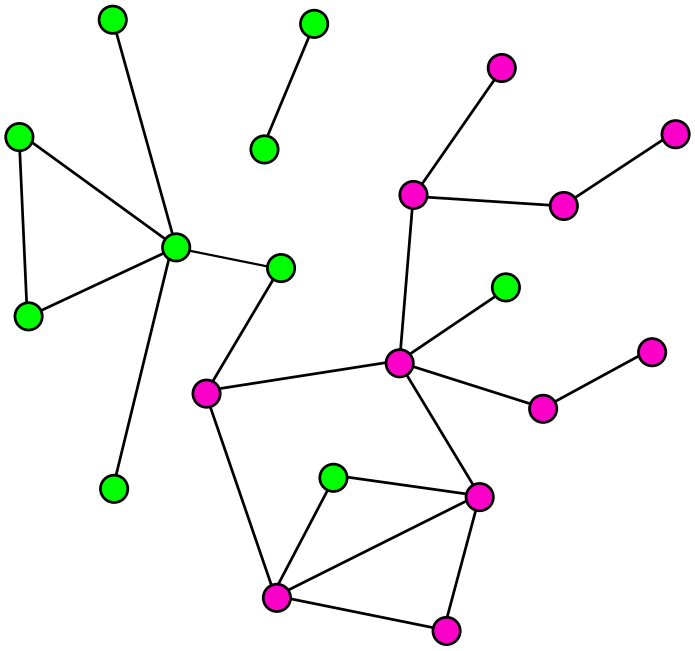

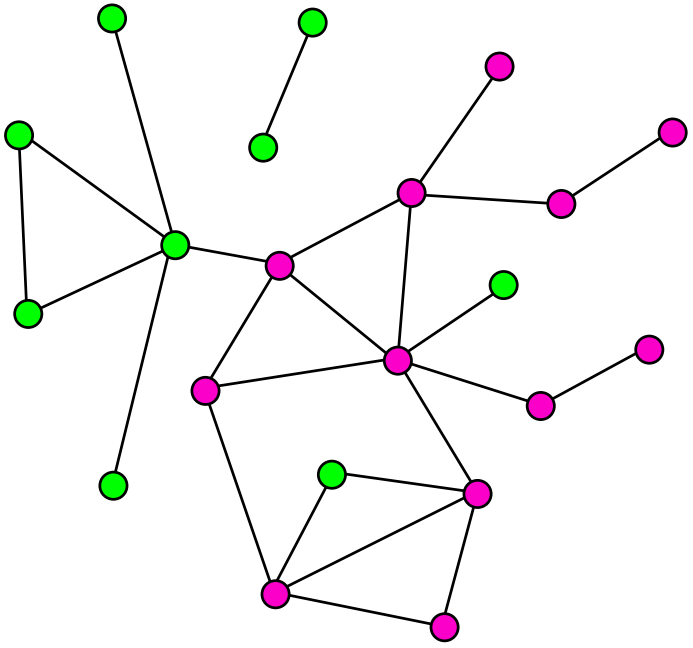

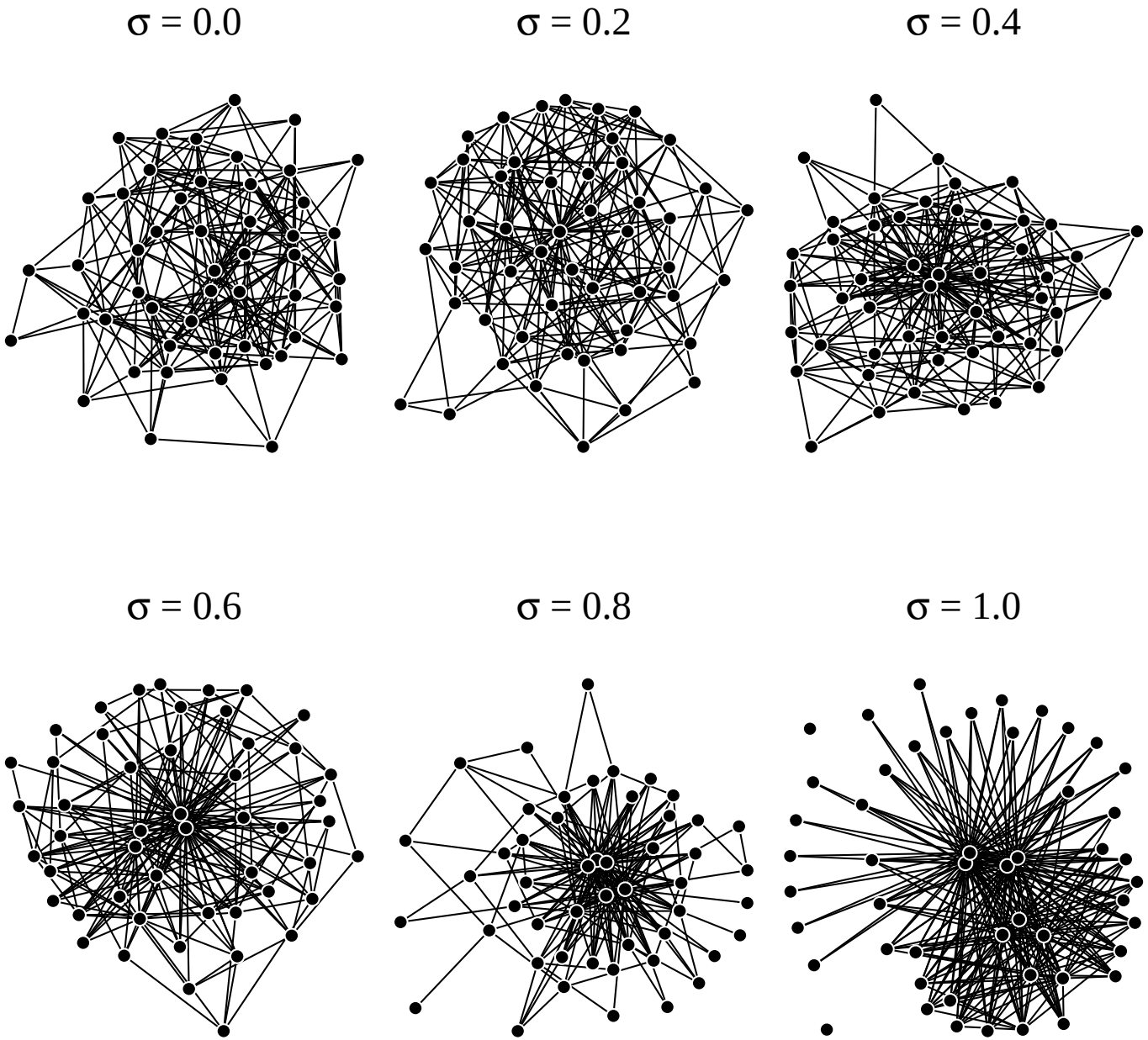
σ = 0.0
σ = 0.2
σ = 0.4
σ = 0.6
σ = 0.8
σ = 1.0

Good behaviour

Does the model give neutral change? If so, what do the trajectories look like?

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Dominance For most time, the language community relaxes into a state in which one variant dominates, so that most or even all speakers use that variant.

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Shifting An innovatory variant ('mutant') can nonetheless begin to propagate and eventually saturate the community; thus, the community may change from dominance by X to dominance by $Y \neq X$.

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Shifting An innovatory variant ('mutant') can nonetheless begin to propagate and eventually saturate the community; thus, the community may change from dominance by X to dominance by $Y \neq X$.

Monotonicity Such invasions proceed in a (quasi)monotone manner with the frequency of the innovation increasing along a smooth propagation curve (*modulo* stochastic noise).

Good behaviour

Quantification:

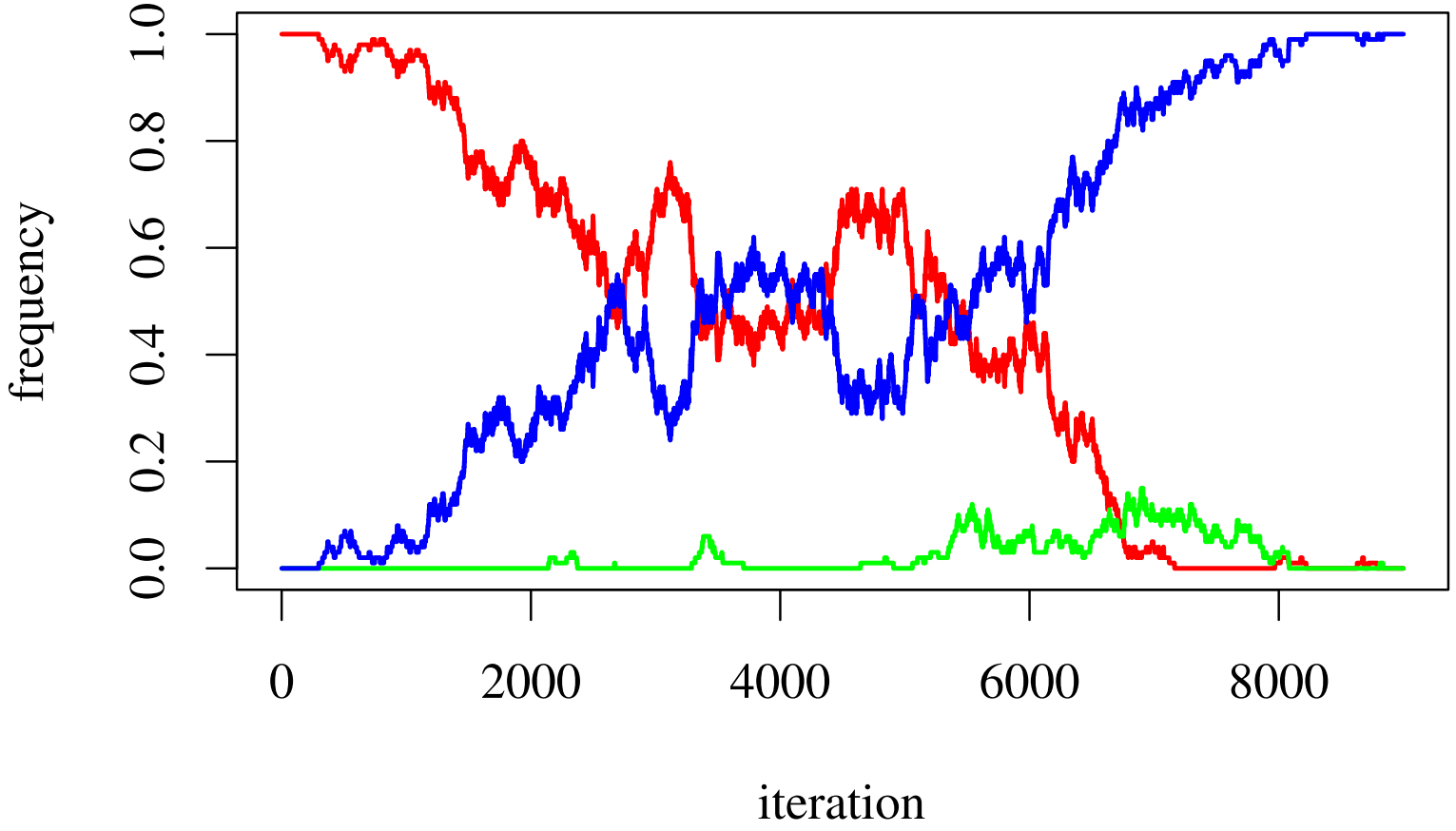
$$D = \left\langle \frac{C}{C-1} \left(\sum_{r=1}^{C} x_r(t)^2 - \frac{1}{C} \right) \right\rangle \tag{1}$$

$$S = \left\langle \begin{cases} 1 \text{ if } |\{r = 1, \ldots, C : x_r(t) = 1 \text{ for some } t\}| \geq 2 \\ 0 \text{ otherwise} \end{cases} \right\rangle \tag{2}$$

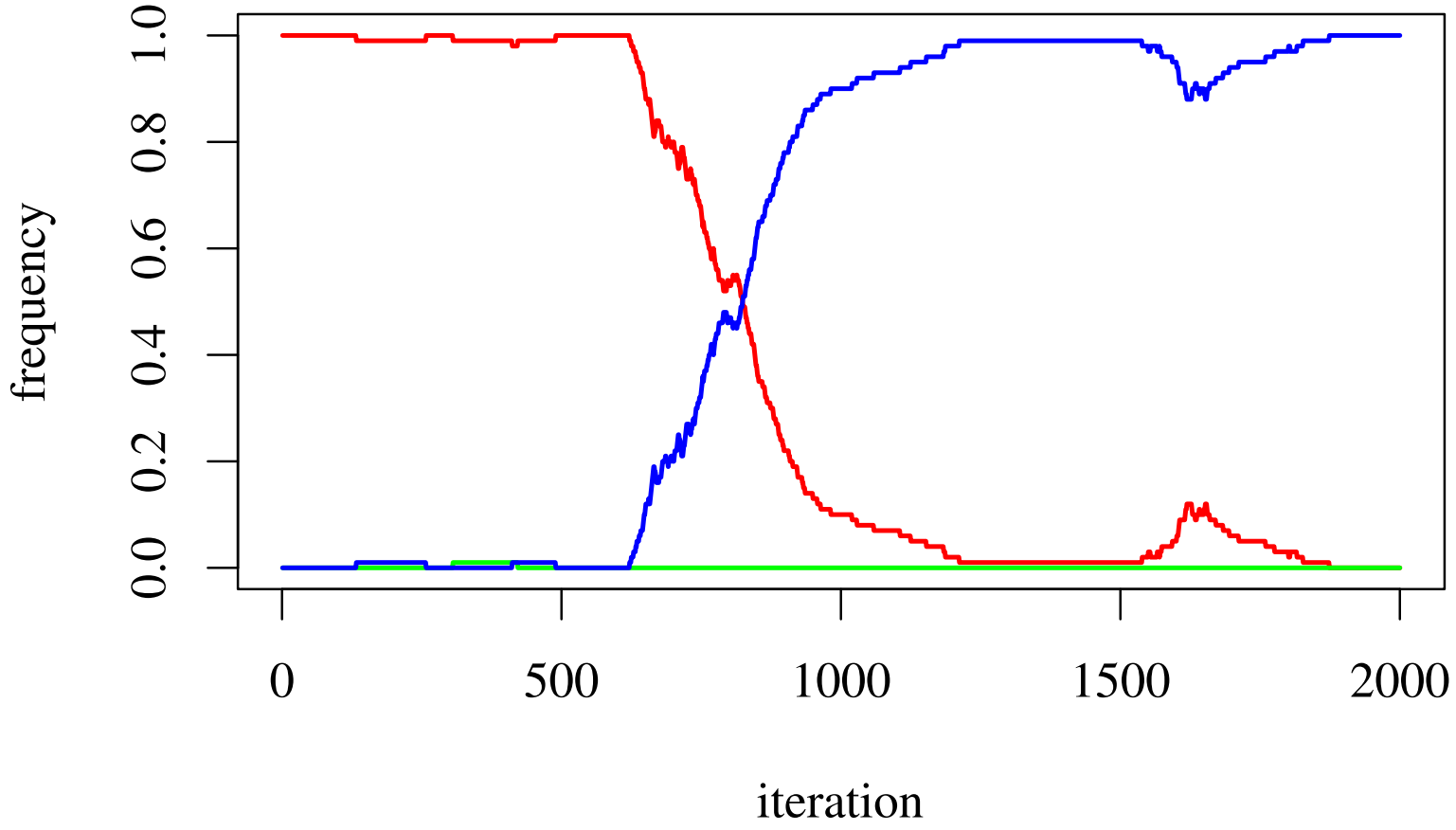
$$M = 1 - \frac{1}{\tau} \left\langle \sum_{r=1}^{C} \sqrt{m^+(r, t_0, \tau) m^-(r, t_0, \tau)} \right\rangle \tag{3}$$

$$W = DSM \tag{4}$$

All range from 0 (ill-behaved) to 1 (well-behaved).

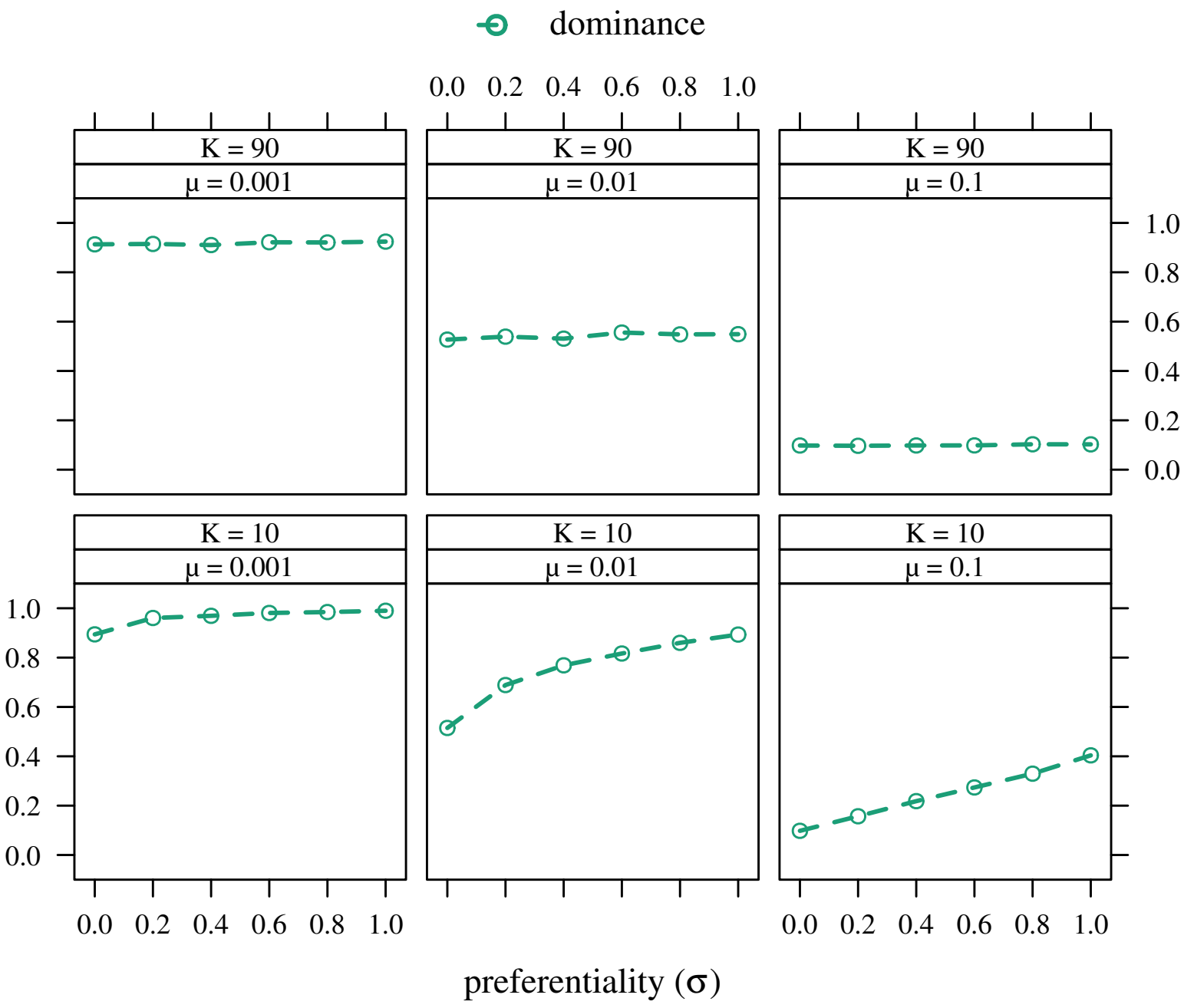


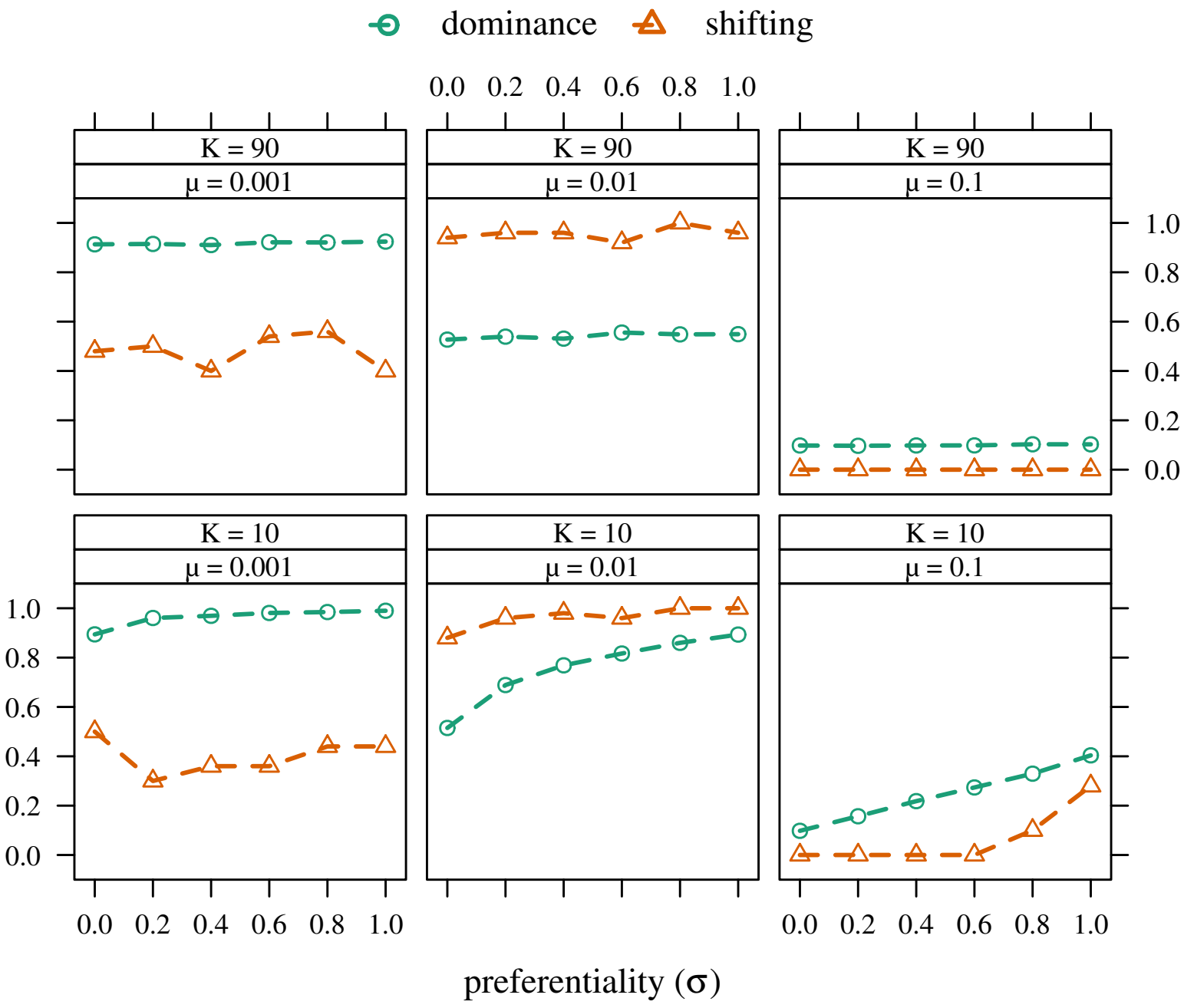
$$D = 0.53, \quad S = 1, \quad M = 0.72$$
$$W = DSM = \mathbf{0.38}$$



$$D = 0.85, \quad S = 1, \quad M = 0.96$$
$$W = DSM = \mathbf{0.82}$$

Results


dominance
K = 90
μ = 0.001
K = 90
μ = 0.01
K = 90
μ = 0.1
K = 10
μ = 0.001
K = 10
μ = 0.01
K = 10
μ = 0.1
0.0
0.2
0.4
0.6
0.8
1.0
preferentiality (σ)


dominance
shifting
K = 90
μ = 0.001
K = 90
μ = 0.01
K = 90
μ = 0.1
K = 10
μ = 0.001
K = 10
μ = 0.01
K = 10
μ = 0.1
0.0
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0.8
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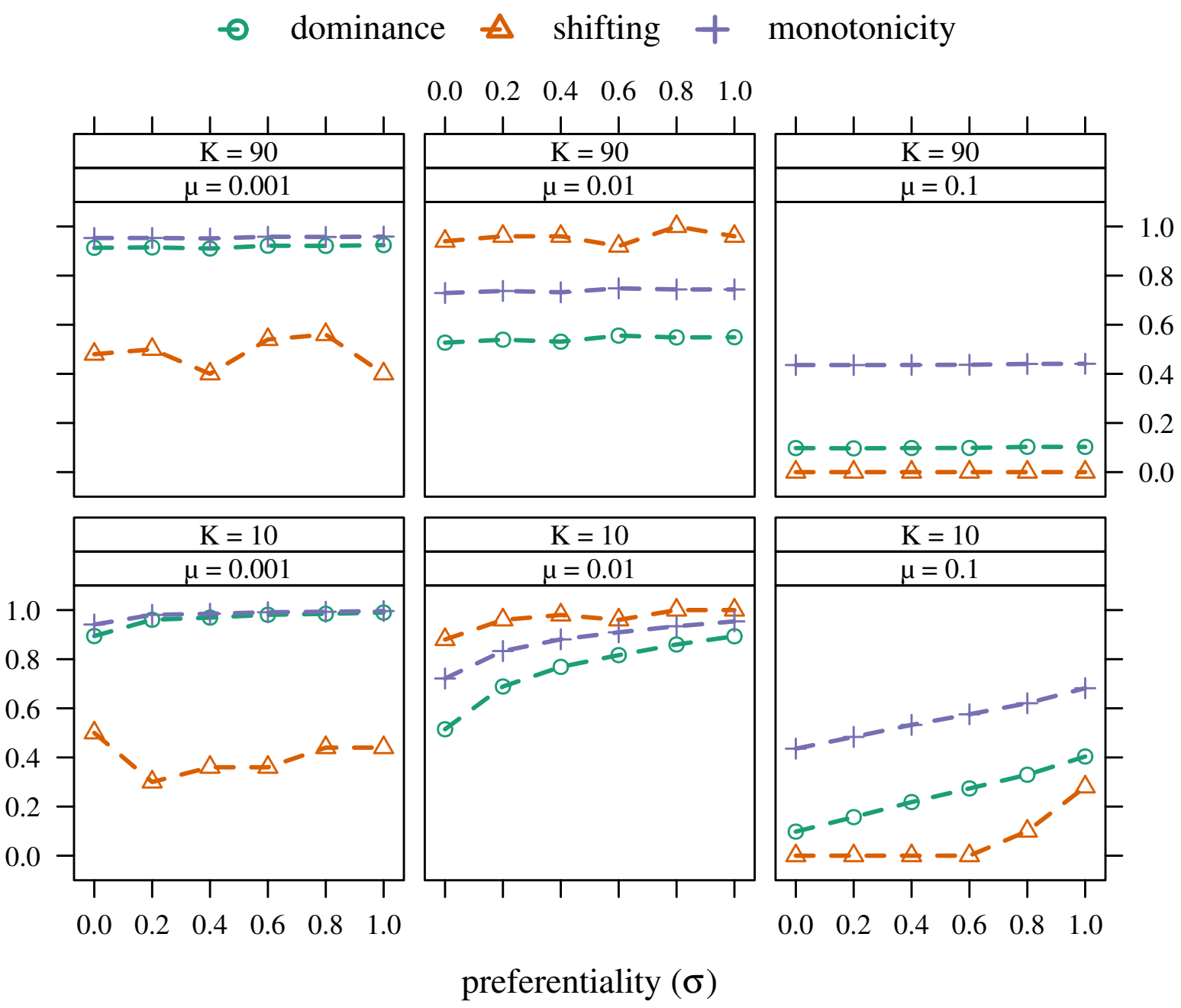
preferentiality (σ)


dominance
shifting
monotonicity
K = 90
μ = 0.001
K = 90
μ = 0.01
K = 90
μ = 0.1
K = 10
μ = 0.001
K = 10
μ = 0.01
K = 10
μ = 0.1
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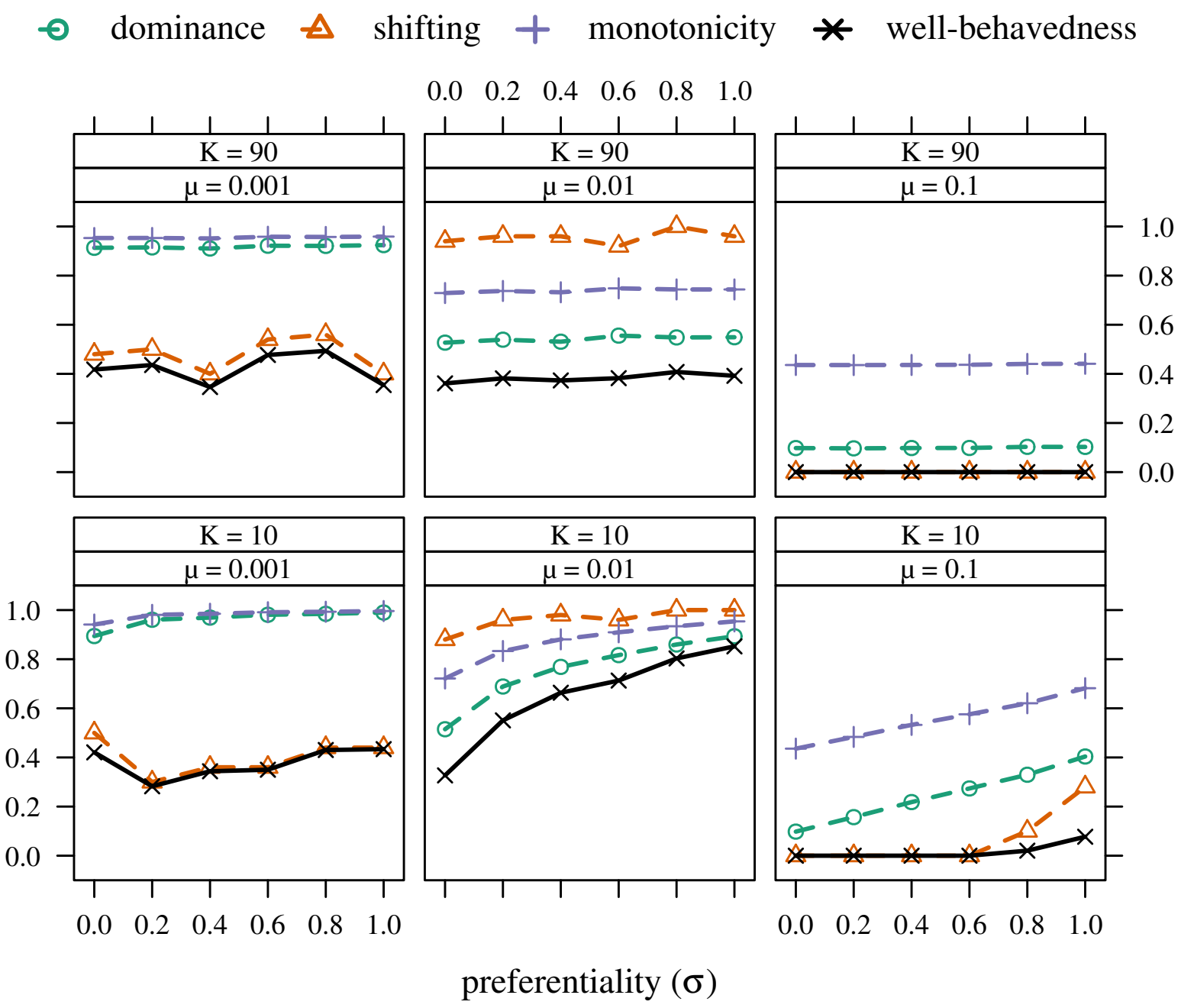
1.0
preferentiality (σ)


dominance
shifting
monotonicity
well-behavedness
K = 90
μ = 0.001
K = 90
μ = 0.01
K = 90
μ = 0.1
K = 10
μ = 0.001
K = 10
μ = 0.01
K = 10
μ = 0.1
0.0
0.2
0.4
0.6
0.8
1.0
preferentiality (σ)

Why neutral change?

- Let 'centre' refer to the set of speakers whose degree is greater than K
- Assume σ is large
- Then the centre acts as both a **conserver** of conventional variants and a **distributor** of innovatory variants
  - As long as the centre is homogeneous, it conserves, since most new speakers are connected to it
  - As soon as a 'mutant' invades the centre, it has a significant probability of propagating, and a cascade ensues
  - For $\sigma = 1$, we can calculate the probability of such a **centre-mutant** propagating their variant to at least one other speaker during their lifetime:

Why neutral change?

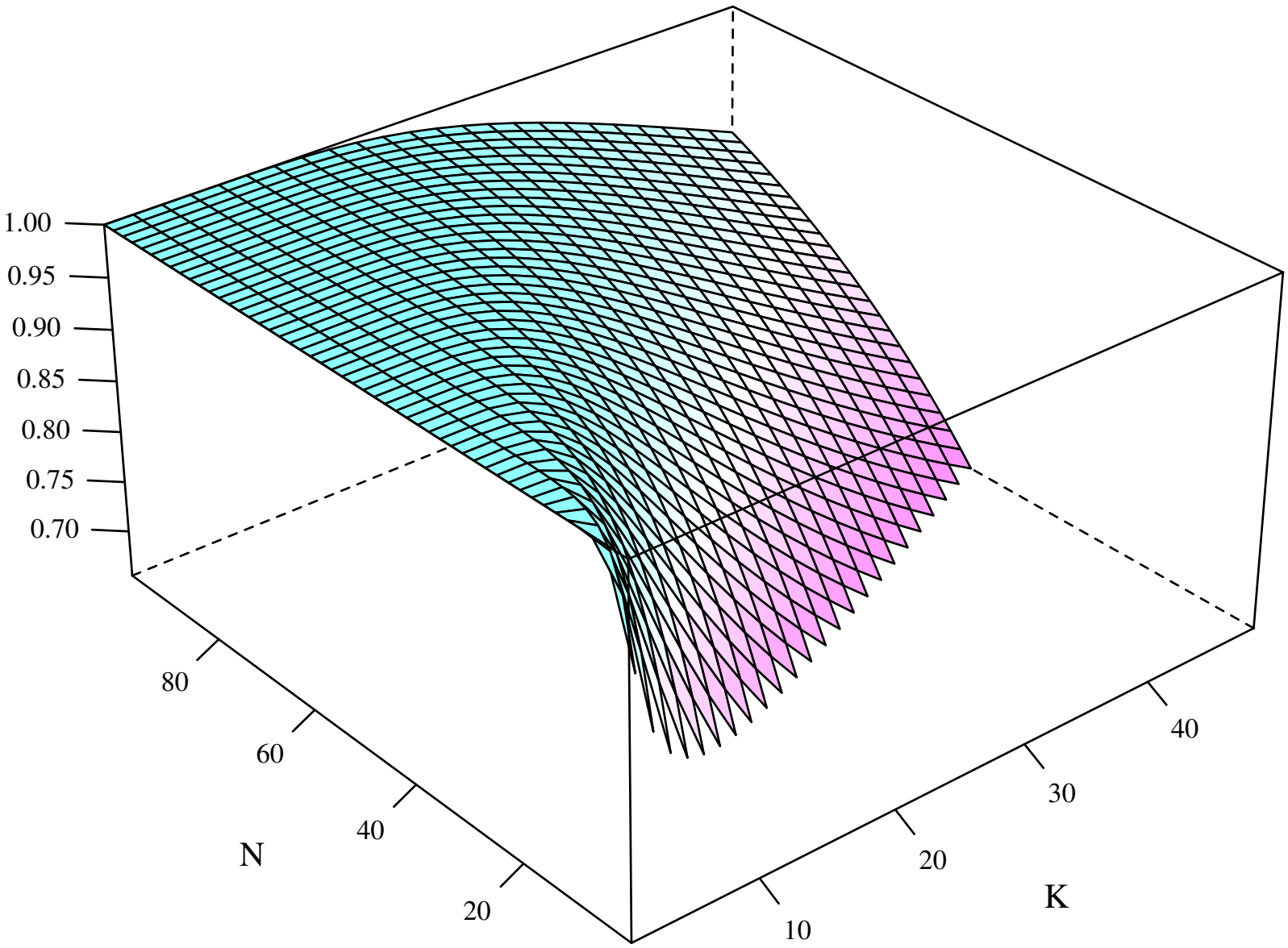
Theorem

Let $\sigma = 1$. Then the probability of a speaker with degree greater than K spreading their variant to at least one other speaker during their lifetime is

$$q = 1 - \left(1 - \frac{1}{K} + \mu\left(\frac{1}{K} - \frac{1}{C}\right)\right)^N;$$

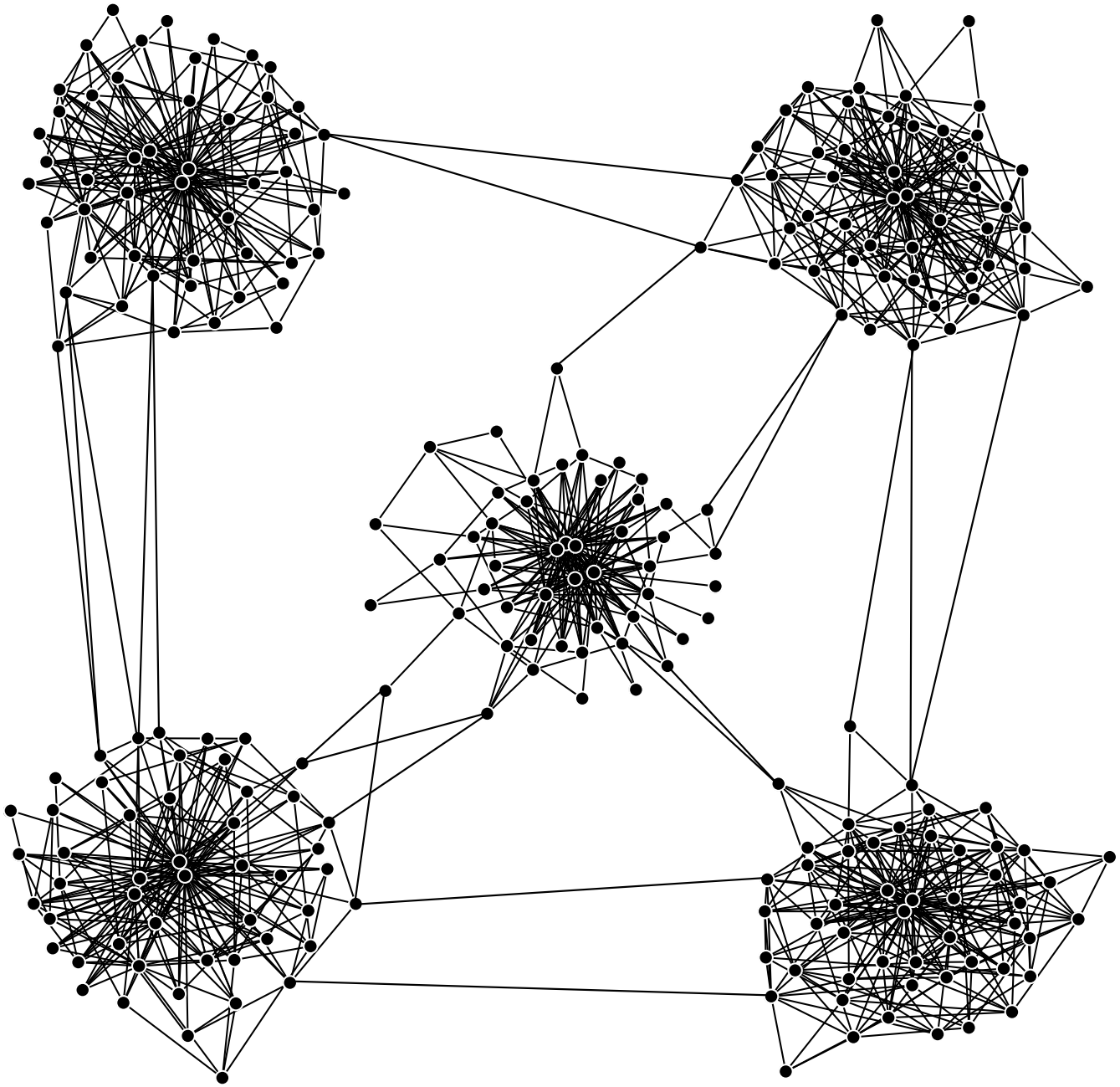
this is ***bounded from below by*** $1 - e^{-1} \approx 0.63$ *for* $\mu < 1/K$.

Moreover, $q \to 1$ ***as*** $K \to 1$ ***and*** $\mu \to 0$.


1.00
0.95
0.90
0.85
0.80
0.75
0.70
80
60
40
20
N
10
20
30
40
K

Interpretation

- Model intended as a simple proof-of-concept framework for exploring the neutrality hypothesis
- However, can fruitfully be interpreted in terms of sociolinguistic theory, although more work is needed on this front
- Milroy and Milroy (1985): the role of **strong (network) ties** vs. **weak ties** in language change
- Proposition: Networks with high σ can be thought of as a formal representation of the Milroyan notion of a **closeknit community**



Interpretation

- Proposition: Networks with high σ can be thought of as a formal representation of the Milroyan notion of a **closeknit community**
- Corollary: Since such networks display neutral change, the notion of **prestige** needs to receive a reassessment.
  - Prestige need not be variant prestige, as often assumed
  - In contrast, prestige may be most fruitfully thought of as residing in the **degree centrality** of individual speakers
  - The results of this research show that when strong **hubs** exist, variant prestige is unnecessary for actuation and propagation of linguistic variants

Conclusion

- In this talk, I have presented results from mathematical and computational work seeking to address the status of neutral theory in diachronic linguistics
- I have established some **sufficient** conditions for (well-behaved) neutral change
- I am not claiming that these conditions are **necessary**
- I am also not claiming that the proposed network structure and rewiring dynamics are particularly realistic
  - They represent the simplest imaginable step from static to dynamic networks that incorporates a notion of clustering
  - A fruitful line for future research is to turn this simple network algorithm into something more realistic, in order to mimic the topology and dynamics of real human social networks – data and techniques exist

References

Blythe, R. A. and Croft, W. (2012). S-curves and the mechanisms of propagation in language change. *Language*, 88(2):269–304.

Blythe, R. A. and McKane, A. J. (2007). Stochastic models of evolution in genetics, ecology and linguistics. *Journal of Statistical Mechanics: Theory and Experiment*, 07:P07018.

Fagyal, Z., Swarup, S., Escobar, A. M., Gasser, L., and Lakkaraju, K. (2010). Centers and peripheries: Network roles in language change. *Lingua*, 120:2061–2079.

Itkonen, E. (1982). Short-term and long-term teleology in linguistic change. In Maher, J. P., Bomhard, A. R., and Koerner, E. F. K., editors, *Papers from the 3rd International Conference on Historical Linguistics*, pages 85–118. Benjamins, Amsterdam.

Ke, J., Gong, T., and Wang, W. S.-Y. (2008). Language change and social networks. *Communications in Computational Physics*, 3(4):935–949.

Labov, W. (1972). *Sociolinguistic patterns*. University of Pennsylvania Press, Philadelphia, PA.

Lass, R. (1997). *Historical linguistics and language change*. Cambridge University Press, Cambridge.

Lightfoot, D. (1979). *Principles of diachronic syntax*. Cambridge University Press, Cambridge.

Milroy, J. and Milroy, L. (1985). Linguistic change, social network and speaker innovation. *Journal of Linguistics*, 21(2):339–384.

Niyogi, P. and Berwick, R. C. (1997). A dynamical systems model for language change. *Complex Systems*, 11:161–204.

Ohala, J. J. (1989). Sound change is drawn from a pool of synchronic variation. In Breivik, L. E. and Jahr, E. H., editors, *Language change: Contributions to the study of its causes*, pages 173–198. Mouton de Gruyter, Berlin.

Pierrehumbert, J. B. (2001). Exemplar dynamics: Word frequency, lenition and contrast. In Bybee, J. L. and Hopper, P. J., editors, *Frequency and the emergence of linguistic structure*, pages 137–157. Benjamins, Amsterdam.

Roberts, I. and Roussou, A. (2003). *Syntactic change: A minimalist approach to grammaticalization*. Cambridge University Press, Cambridge.

Trudgill, P. (2008). Colonial dialect contact in the history of European languages: On the irrelevance of identity to new-dialect formation. *Language in Society*, 37:241–280.

Vennemann, T. (1993). Language change as language improvement. In Jones, C., editor, *Historical linguistics: Problems and perspectives*, pages 319–344. Longman, London.

Yang, C. D. (2000). Internal and external forces in language change. *Language Variation and Change*, 12:231–250.

Thank you!

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